

Flagging Emotional Manipulation: Impacts of Manipulative Content Warnings on Sharing Intentions and Perceptions of Health-Related Social Media Posts

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Abstract

Misinformation and manipulative online content pose significant threats to public health and decision-making. The rise of AI-generated content and reduced reliance on fact-checking programs by platforms have exacerbated these challenges. Although fully automated detection of misinformation remains beyond current capabilities, researchers have developed effective methods for identifying linguistic features, such as emotionally manipulative language, that are associated with misleading content. However, little is known about how warning labels for manipulative content influence user perceptions and behavior. In an experimental survey of 945 Americans, we investigated the effects of manipulative content warning labels on responses to true and false health-related social media posts. Our findings highlight differences in user reactions to posts with various types of labels, with implications for mitigating the impact of manipulative content. We offer recommendations for the design of effective labeling systems and future research directions to enhance online content moderation strategies.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**;
Empirical studies in collaborative and social computing.

Keywords

health misinformation, manipulative rhetoric, persuasive writing strategy, warning labels, social media

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1 Introduction

Online health misinformation is a crucial challenge, potentially fostering conspiracy beliefs and poor healthcare decisions [8]. Common definitions distinguish between disinformation – intentionally

false statements; misinformation – false statements that may be unintentional; and malinformation – where accurate information is presented out of context to make a misleading point [6]. Fact-check labels are a common and broadly effective strategy for reducing the belief and sharing of misinformation [35]. However, fact-checking is laborious and time-consuming [36], and the increasing use of generative AI to create “synthetic” misinformation exacerbates these challenges [12, 45]. These scalability challenges are one reason platforms have shifted away from fact-check organizations to rely on community notes [33, 41]. However, some evidence suggests community notes may not effectively curb engagement with misinformation [10].

The same AI advancements that contribute to synthetic misinformation may aid in automating fact-checking. While fully automated fact-checking remains beyond current capabilities [19], AI demonstrates proficiency in evaluating features of a text associated with misinformation, such as persuasive techniques [9, 25, 40] and emotional sentiment [1, 7]. Presenting evidence of suspicious cues to end-users has been shown to improve their ability to detect harmful content in contexts like phishing scams [2, 3]. However, despite proposals to present persuasive rhetoric warnings to users to aid misinformation detection [25], the impacts of such warnings on user engagement with misinformation remains underexplored.

To address this gap, we surveyed 945 American social media users to evaluate the effects of warning labels about manipulative text on perceptions of posts’ accuracy and manipulativeness, as well as sharing intention. Specifically, we evaluated warning labels about attempts to manipulate readers’ emotions since this is a common misinformation technique [16]. To understand the effects of the warning labels in multiple conditions, the study included both true and false posts, as well as posts containing or not containing emotionally manipulative text. Our study compared two types of manipulative content warnings against a control condition with no warning labels and against conventional fact-checking labels.

Our research questions were:

RQ1: To what extent do fact-check and manipulative content warnings affect perceptions of a post’s (a) accuracy, (b) manipulativeness, and (c) intentions to share that post?

RQ2: How do participants evaluate the different types of labels?

Our findings suggest that manipulative content warnings influenced beliefs about manipulative posts but had a stronger effect on true + manipulative posts than false + manipulative ones. The results do not demonstrate that these manipulative content warnings could curb engagement with misinformation on their own, but they highlight potential applications for raising awareness of

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manipulative rhetorical techniques and opportunities to enhance these labels with additional information.

This paper contributes to knowledge about the effectiveness of warning labels about emotionally manipulative content on mitigating misinformation, particularly in contexts where fact-checking is infeasible. Based on the results, we discuss opportunities for future research and designs.

2 Related Work

2.1 Emotional manipulation and misinformation

Research suggests that reliance on emotions over reason increases susceptibility to fake news [5, 34]. Thus, it is no surprise that emotional arguments and rhetoric appear more frequently in online health misinformation than factual content [37]. Emotional appeals have been identified as a core feature of propaganda [16]. However, it is important to acknowledge that individual susceptibility varies, and there are cases when emotional content decreases rather than bolsters credibility judgments of online information [26, 39]. Still, a strong correlation between belief in emotionally manipulative content and belief in false information [32] highlights its importance as a driver of misinformation.

Detecting emotionally manipulative text. Extensive research has developed techniques for detecting persuasive or manipulative text [20]. For example, Huffaker et al. [22] used crowdsourcing to detect emotionally manipulative language, such as warnings one should follow a specific action “or else” (p. 1) and Ghanem et al. [15] demonstrated that a text’s emotional signals can be used to assess its credibility. Similarly, Kamali et al. [25] demonstrated that identifying persuasive strategies, such as appeals to emotion, can improve the performance of misinformation detection models. These studies highlight the potential of automated systems to flag manipulative rhetorical styles.

These style-based approaches, which analyze manipulative techniques within a text itself, have the advantage of not requiring external fact-check information [20]. A key implication is that these approaches may be applicable in contexts where misinformation spreads faster than fact-checkers can respond. Further, based on insights that awareness of persuasive tactics can reduce susceptibility to persuasion attempts [17], researchers have suggested that warning labels shown to platform users could assist individuals in misinformation detection [25].

2.2 Misinformation warning labels

A wide variety of interventions have been explored to reduce belief in and sharing of misinformation [28]. Preemptive approaches, such as prebunking and inoculation, train users to recognize common misinformation strategies, improving their ability to detect manipulation [42] and reducing engagement with false information [30]. However, these approaches are typically implemented prior to exposure rather than in situ.

In-situ interventions, such as warning labels integrated into social media feeds, aim to address misinformation at the moment of exposure. These include fact-check labels and community notes,

which have been deployed on social media platforms to flag disputed content. HCI research has shown that the effects of such labels vary substantially based on their content and design [14]. For example, Lim and Perrault [31] found that generic warnings reduced intentions to comment or share posts but not to like them, and that more specific labels had inconsistent effects. Jia et al. [24] found that responses to warning labels can vary between liberal and conservative users. Warning labels also must consider usability and accessibility. High friction warnings that delay or prevent access to content are effective but are disliked by some users for interrupting their browsing [18]. On the other hand, poor accessibility may prevent some users from noticing labels altogether [43].

Warnings about emotionally manipulative content may require particular care since manipulateness does not ensure falsehood. Heavy-handed warnings could result in frustrations and perceived unfairness [11]. Thus, there is a possibility that warning labels about *potentially, but not assuredly* false or manipulative content could be met with resistance.

Despite progress in detecting manipulative text and the potential for warning labels about manipulation to mitigate misinformation, there is limited empirical research on how such labels affect user perceptions and behaviors. This study addresses this gap by investigating the impact of warnings about emotionally manipulative content and exploring their potential to support users in evaluating online information.

3 Method

Participants viewed eight health-related social media posts that varied in terms of factual accuracy and emotional manipulation and provided ratings of accuracy, manipulateness, and sharing intention using 7-point ordinal responses. There were four conditions with different warning interfaces, which were used to compare two types of manipulative text warning labels to a control condition (no warnings) and to more conventional fact-check warning labels. After responding to the social media posts, participants evaluated the warning labels and described their social media practices, attitudes about information, and demographics. After the study, participants were shown debriefing information explaining the research’s purpose and informing them about which posts were accurate, according to fact-checkers.

The average study duration was 8 minutes 55 seconds, and participants were paid £2.50 (about \$3.17 USD).¹ This study was approved by our institution’s ethics review board.

3.1 Recruitment and participants

We recruited 1000 Participants using Prolific² with the following eligibility criteria: Resident of United States, fluent in English; approval rate of 98-100%; equal percentage of male and female participants. There were two attention check questions, and 55 participants who failed an attention check were excluded, leaving 945 included in the study. Participants age, race, and household income are reported in Table 1.

¹Participants were paid in GBP because Prolific is based in the United Kingdom.

²<https://www.prolific.com/>

Table 1: Participant demographics

Age		Race	Household income		
18-24	134 (14%)	Asian	48 (6%)	Less than \$10,000	32 (4%)
25-34	320 (34%)	Black or African American	205 (24%)	\$10,000 to \$24,999	74 (8%)
35-44	233 (25%)	Hispanic or Latino	38 (4%)	\$25,000 to \$49,999	179 (20%)
45-54	165 (17%)	Middle Eastern or North African	1 (0%)	\$50,000 to \$74,999	188 (21%)
55-64	56 (6%)	Native American or Alaska Native	4 (0%)	\$75,000 to \$99,999	151 (17%)
65-74	33 (3%)	White	568 (66%)	\$100,000 to \$149,999	190 (21%)
75+	4 (0%)	Multiple races	74 (9%)	\$150,000 to \$249,999	74 (8%)
		No answer	5 (1%)	\$250,000 or more	31 (3%)
				No answer	26 (3%)

3.2 Study design

The eight social media posts displayed to participants were selected from a dataset used by Kamali et al. [25], which is a health-focused subset of the MultiFC dataset [4], which includes fact-check claims collected from 26 fact-checking websites. The dataset also included annotations about each post’s persuasive rhetorical strategies, classified using an automated Kamali et al.’s system [25]. Of the persuasive rhetorical strategies, we focused on *emotional appeals*—texts with language designed to evoke fear, anger, hope, or anxiety. Our motivation for using Kamali et al.’s annotations was to represent current capabilities for detecting persuasive strategies based on short text passages. For example, one post asserted that medical researchers are “*super excited*” to have discovered Dandelion Root’s “*potential to cure cancer!*” This post had been fact-checked as “Mostly false”³ and was annotated as containing an appeal to emotion (hope). Therefore, this was classified as *false + manipulative*.

Post types: Based on the above annotations, participants were shown two posts of each of the following types: *True + not manipulative*; *true + manipulative*; *false + not manipulative*; and *false + manipulative*. To control for order effects, the posts were shown in random order for each participant.

Experimental conditions: Participants were grouped into four conditions, which were shown different types of warning labels. (1) *Control group* (n = 237): Posts were displayed with no warning labels; (2) *Fact-check warning* (n = 241): False posts were displayed with a warning stating, “This post may contain misleading or false information. Before sharing, consider whether the poster is trying to influence your beliefs.”; (3) *Manipulative content warning* (n = 235): Posts containing *emotional appeals* were displayed with a warning stating, “This post contains language that may be trying to manipulate your emotions. Before sharing, consider whether the poster is trying to influence your beliefs.”; (4) *Manipulative content warning + explanation* (n = 232): Posts with emotional appeals had the same warning as condition (3), plus an LLM ChatGPT-generated explanation of why the text was determined to be manipulative. The purpose of condition (4) is to explore the effects of tailored explanations created through generative AI, which have been proposed in several studies about misinformation interventions [13, 21, 27]. Warning label designs and the full text shown to participants, including manipulation explanations for condition

(4), are in Appendix A.1. The ChatGPT prompt for condition (4) is described in Appendix A.2.

Responses to each post: After viewing each post, participants evaluated its accuracy and their intention to share the post using questions adapted from Pennycook and Rand [38], as well as the post’s manipulateness using questions adapted from Saleh et al. [42]. Questions were rated on a 7-point scale from “strongly agree” to “strongly disagree.”

Warning label evaluations: Participants in groups 2, 3, and 4 were asked to evaluate how *trustworthy*, *helpful*, and *annoying* the labels were, using a 7-point scale from “strongly disagree” to “strongly agree.”

3.3 Data analysis

3.3.1 Baseline equivalence checks. We conducted baseline equivalence checks to identify any between-group differences that could affect our variables of interest. Using Kruskal-Wallis H-tests, no statistically significant differences were detected regarding income, education, political orientation, or frequency of viewing or sharing social media posts in general. However, there was a significant between-group difference in age, $H(3) = 8.72$, $p = .033$, with a post-hoc Dunn’s test showing that Group 2 was slightly younger than Group 1 ($p = .021$). To account for this, age is included in our models as a control variable.

3.3.2 Regression models. To address RQ1, we used three mixed-effects ordinal logistic regression models. The dependent variables for the three models were ordinal indicators of perceived accuracy, perceived manipulateness, and sharing intention. The independent variables were post-type and experimental conditions. To evaluate the effects of warning labels on different types of posts, the models included interaction terms between post type and experimental condition. Age was included as a control variable. Since each participant rated multiple posts, we accounted for the non-independence of observations by including random intercepts for participants. This allowed the model to account for individual differences in baseline ratings across posts.

4 Results

Table 2 presents the results of mixed-effects ordinal logistic regression models addressing RQ1.

³<https://www.snopes.com/fact-check/dandelion-kills-cancer/>

Table 2: Mixed-Effects Ordinal Logistic Regression models predicting perceived accuracy, perceived manipulateness, and sharing intention. Standard error in parentheses. * $p < .05$, ** $p < .01$, * $p < .001$**

	Perceived Accuracy		Perceived manipulateness		Share intention	
Post type (<i>reference group = True + Non-manip.</i>)						
True + Manip.	0.403***	(0.12)	-0.324**	(0.12)	0.306*	(0.14)
False + Non-manip.	-0.311**	(0.12)	0.386***	(0.12)	0.203	(0.13)
False + Manip.	-1.317***	(0.12)	1.048***	(0.12)	-0.538***	(0.14)
Condition (<i>reference group = Control</i>)						
(2) Fact-check warning	-0.375*	(0.17)	-0.057	(0.16)	-1.189***	(0.29)
(3) Manip. content warning	0.053	(0.17)	0.150	(0.16)	0.073	(0.28)
(4) Manip. content warning + explanation	-0.141	(0.17)	0.023	(0.17)	-0.663*	(0.29)
Interaction with condition = (2) "Fact check warning"						
True + Manip.	0.129	(0.17)	-0.089	(0.16)	0.095	(0.20)
False + Non-manip.	-1.060***	(0.17)	0.910***	(0.16)	-1.017***	(0.20)
False + Manip.	-0.787***	(0.17)	0.632***	(0.16)	-0.492*	(0.20)
Interaction with condition = (3) "Manipulative content warning"						
True + Manip.	-0.486**	(0.17)	0.507**	(0.16)	-0.519**	(0.19)
False + Non-manip.	0.017	(0.17)	-0.145	(0.16)	-0.202	(0.19)
False + Manip.	-0.089	(0.17)	0.024	(0.17)	-0.263	(0.19)
Interaction with condition = (4) "Manipulative content warning + explanation"						
True + Manip.	-0.183	(0.17)	0.695***	(0.17)	-0.063	(0.20)
False + Non-manip.	-0.014	(0.17)	-0.002	(0.17)	-0.032	(0.19)
False + Manip.	0.048	(0.17)	0.390*	(0.17)	0.134	(0.20)
Demographic control						
Age	-0.181***	(0.04)	0.054	(0.04)	-0.354***	(0.07)
N objects = 7560; N groups = 945						

First, older ages were significantly associated with slightly lower perceived accuracy ($\beta = -0.181$, $p = .000$) and sharing intention ($\beta = -0.354$, $p = .000$) in general.

The results under *Post type* indicate how participants across all condition groups responded to different types of posts. Compared to the reference group of True + Non-manipulative posts, participants viewed True + Manipulative posts as more accurate ($\beta = 0.403$, $p = .001$) and less manipulative ($\beta = -0.324$, $p = .005$), and had a slightly higher sharing intention ($\beta = 0.306$, $p = .024$). This suggests that, in general, the manipulateness of true posts was effective.

On the other hand, participants in all conditions had negative evaluations of false posts. False + Non-manipulative posts were viewed as slightly less accurate ($\beta = -0.311$, $p = .008$) and slightly more manipulative ($\beta = 0.386$, $p = .001$) compared to the reference group, though there was no significant difference in sharing intention. False + Manipulative posts were responded to very negatively, as significantly less accurate ($\beta = -1.317$, $p = .000$) and more manipulative ($\beta = 1.048$, $p = .000$), and with lower sharing intention

($\beta = -0.538$, $p = .000$). This suggests that participants were generally good at discerning false posts, especially manipulative ones.

Condition = Fact-check warning: Results under the *Condition* heading indicate how responses varied across groups, irrespective of post type. Participants in the "Fact-check warning" group had lower accuracy ($\beta = -0.375$, $p = .027$) and sharing intention ($\beta = -1.189$, $p = .000$) ratings toward all posts. Results under the heading *Interaction with condition = "Fact check warning"* reveal further details. Specifically, on top of the generally negative accuracy and sharing intention ratings that members of this condition had toward all posts, they were significantly more likely than others to react negatively to false posts. Effect sizes are smaller for False + Manipulative posts (Accuracy $\beta = -0.787$, $p = .000$; Manipulateness $\beta = 0.632$, $p = .000$; Sharing intention $\beta = -0.492$, $p = .016$) than for the False + Non-manipulative posts (Accuracy $\beta = -1.060$, $p = .000$; Manipulateness $\beta = 0.910$, $p = .000$; Sharing intention $\beta = -1.017$, $p = .000$). This may be because participants already had quite negative evaluations of False + Manipulative posts, irrespective of study

condition. In combination, these results reveal that fact-check labels were effective at reducing engagement with false posts but also reduced accuracy and sharing intention for true posts, to a lesser extent.

Condition = Manipulation warning: Manipulation warnings resulted in lower accuracy ratings ($\beta = -.486$, $p = .004$), higher perceived manipulateness ($\beta = 0.507$, $p = .002$), and lower sharing intention ($\beta = -0.519$, $p = .007$) for True + Manipulative posts, compared to the reference group of True + Non-manipulative posts. The effect sizes for this reduction in confidence are slightly larger than the opposite coefficients that True + Manipulative posts had in general (under the *Post type* subheading). This suggests that the manipulation warnings more or less eliminated the advantage that manipulative true posts had over non-manipulative true posts.

Condition = Manipulation warning + explanation: In this condition, warnings about emotional manipulation were accompanied by a ChatGPT-generated explanation about the nature of the manipulation. Results under the *Condition* subheading indicate that these labels were associated with lower sharing intention in general ($\beta = -0.663$, $p = .021$). There were no significant interactions between this condition and perceived accuracy or sharing intention of specific post types. However, this condition was associated with higher perceived manipulateness for both true ($\beta = 0.695$, $p = .000$) and false ($\beta = 0.390$, $p = .020$) manipulative posts. This suggests that the explanations in this condition increased participants' ability to evaluate manipulateness in line with the warning content. However, this did not correspond to changes in behavior or accuracy assessment.

Participant evaluations of the warnings: We addressed RQ2 using Kruskal-Wallis H-tests with Bonferroni-adjusted p-values, which found no significant differences in ratings of trustworthiness ($Mdn = 5$, $\chi^2(2) = 4.391$, adjusted $p = .334$) or annoyance ($Mdn = 5$, $\chi^2(2) = 4.988$, adjusted $p = .248$). However, there was a significant difference in helpfulness ratings across groups ($\chi^2(2) = 9.060$, adjusted $p = .028$). Post-hoc Dunn tests with Bonferroni correction revealed that fact-check warnings ($Mdn = 2$) were rated as less helpful than manipulative content warnings ($Mdn = 3$, adjusted $p = .04$) and manipulative content warnings with explanations ($Mdn = 3$, adjusted $p = .005$). Ratings of 2 and 3 out of 7 indicate that participants generally disagreed that the labels were helpful, with fact-check labels perceived as the least helpful.

5 Discussion

5.1 Summary of results

This study investigated how fact-check and manipulative content warnings influence user perceptions and behaviors toward health-related social media posts.

Fact-check warnings effectively reduced perceived accuracy and sharing intention, particularly for false content, supporting prior findings on their efficacy in combating misinformation [35]. However, they also slightly lowered perceptions of accuracy and sharing intention for true posts, indicating potential overgeneralization effects.

Manipulative content warnings without explanations (condition 3) neutralized the persuasive advantage of emotional manipulation

for true posts, suggesting a way to overcome the apparent enhancement to credibility that emotional manipulation can furnish [16]. However, these warnings did not significantly affect user responses to false manipulative content, suggesting limited efficacy against outright falsehoods. Warnings with explanatory labels (condition 4) increased participants' recognition of manipulateness in both true and false manipulative content but did not significantly impact accuracy judgments or sharing intentions for false or manipulative posts. Instead, they were associated with lower sharing intention regardless of post type. These findings suggest that while explanations may enhance critical evaluation, they do not directly influence user behavior regarding false or manipulative content.

Finally, participants rated manipulative content warnings as slightly more helpful than fact-check warnings, although helpfulness ratings were generally low. Trustworthiness and annoyance ratings were consistent across conditions.

5.2 Implications and future directions

The results contribute to ongoing discussions about the design of effective interventions for mitigating the spread of misinformation. Prior work [25] suggested that warnings about manipulative rhetorical styles could enhance users' ability to evaluate misinformation. Our results showed that such labels could improve recognition of manipulation, particularly in true posts, but suggest that these labels alone are insufficient to curb misinformation behaviors.

Future research could explore the effects of combining warning labels about emotional manipulation with other information. Combining these warnings with fact-check labels could target false + manipulative posts more precisely. Still, a chief motivation for this study is that fact-check information is not always available. In the absence of fact-check information, more detailed information about manipulative rhetoric [9, 25] or other credibility indicators [44] may provide more effective guidance to users.

Since condition (4)'s warnings with explanations did not impact accuracy beliefs or sharing intentions, they are unlikely to be effective for immediate behavioral changes. However, their impact on perceived manipulateness may make them effective in inoculation strategies, in which education about manipulative tactics can build long-term resistance to those techniques [42]. Future studies could investigate the effect of such warnings over a longer duration or as part of broader inoculation programs.

Additionally, the relatively low helpfulness ratings across all warning labels suggest a need for more tailored interventions. Personalized AI, which could improve warning labels' helpfulness by adapting them to users' blind spots [23].

Finally, this study was conducted in a controlled web survey setting with limited post diversity. Given the apparently higher effectiveness of manipulative content labels on true posts compared to false posts, examining their effects on malinformation or posts with an unclear truth status would be of great value. This could be accomplished by using a larger library of posts or by testing warnings with real-world content using a browser plugin or platform API. In doing so, a benefit of focusing on rhetorical strategies contained within a text itself is that external fact-check or other information is not required. Therefore, the interventions explored

in this study could be extended to contexts like private messaging [29], which are often inaccessible to fact-checkers. Expanding the range of content would provide a more nuanced understanding of how manipulative content warnings perform in varied contexts.

6 Conclusion

This study examined the effects of warnings about emotionally manipulative content on user perceptions and behaviors regarding health-related social media posts in comparison to a control condition with no warnings and compared to conventional fact-check warnings. The results indicated that fact-check warnings were generally effective at reducing engagement with false posts but also slightly reduced engagement with true posts. By contrast, manipulative content warnings reduced perceived accuracy and sharing intention toward true + manipulative posts but did not have a significant effect on false posts. Adding explanations to those warnings led to increased recognition of manipulative strategies but no significant impacts on perceived accuracy or sharing behavior. These findings suggest that manipulative content warnings alone are insufficient to reduce engagement with misinformation. However, they highlight opportunities for future improvements, such as combining emotional manipulation warnings with other credibility indicators and investigating effects on content with a more ambiguous truth status. Based on these results, we propose directions for future research and design interventions aimed at improving user engagement with misinformation.

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A Appendix

A.1 Post contents and warning labels

Figure 1 shows images of the warning labels as they were displayed to participants in each condition.

A.1.1 Post content. Below, we list the text headline and text summary of each of the posts displayed to participants, along with an explanation of the post type (e.g., “True + not-manipulative”), which warning labels were attached to this post in each condition, and the explanatory text that generated for condition (4) “Manipulative content warning + explanation.” “Post types” are based on the criteria defined in section 3.2.

Headline: Study: Manuka honey kills more bacteria than all available antibiotics

Summary: Not all honey is created equal. While the benefits of raw, unprocessed honey have been well-documented over the centuries, Australian researchers have found one type of honey, called Manuka honey, to be better than all known antibiotics.

Post type: False + not manipulative.

Warning labels: Condition (2): Fact-check warning.

Headline: TIP OF THE DAY: WATCH FOR CRABS!

Summary: When buying crabs in the market look for injection marks such as these when buying crabs imported from China. Chinese inject the crabs with formulin to keep them fresh but it is a highly toxic substance. I heard this today at a Thai market.

Post type: False + not manipulative.

Warning labels: Condition (2): Fact-check warning.

Headline: Scientists Find Root That Kills 98% Of Cancer Cells In Only 48 Hours

Summary: Dandelion has been used medicinally since ancient times for its various health benefits. However, the most powerful benefit to come out of this common weed is something that medical researchers are super excited to have “discovered” - which is its potential to cure cancer!

Post type: False + manipulative. (*appeal to emotion: “hope”*)

Warning labels: Condition (2): Fact-check warning. Conditions (3) & (4): Manipulative content warning.

Explanation text for condition (4): “This post contains language that may be intended to make you feel **hope** by presenting dandelion root extract as a potentially revolutionary treatment for cancer, which could be seen as a more promising and less harmful alternative to traditional therapies.”

**Warning label for condition (2):
Fact-check warning**

⚠ This post may contain misleading or false information. Before sharing, consider whether the poster is trying to influence your beliefs.

**Warning label for condition (3):
Manipulative content warning**

⚠ This post contains language that may be trying to manipulate your emotions. Before sharing, consider whether the poster is trying to influence your beliefs.

**Warning label for condition (4):
Manipulative content warning + explanation**

⚠ This post contains language that may be trying to manipulate your emotions. Before sharing, consider whether the poster is trying to influence your beliefs.

This post contains language that may be intended to make you feel **hope** by presenting dandelion root extract as a potentially revolutionary treatment for cancer, which could be seen as a more promising and less harmful alternative to traditional therapies.

Figure 1: Warning label designs for each study condition. Explanatory text in the condition (4) label is an example. See Appendix A.1 for the full text of each post and the matching explanatory content.

Headline: Researchers find a simple cure for cancer, but major pharmaceutical companies are not interested.

Summary: Researchers at the University of Alberta, in Edmonton, Canada have cured cancer last week, yet there is a little ripple in the news or in TV. It is a simple technique using very basic drug. The method employs dichloroacetate, which is currently used to treat metabolic disorders. So, there is no concern of side effects or about their long term effects.

Post type: False + manipulative. (*appeal to emotion: "hope"*)

Warning labels: Condition (2): Fact-check warning. Conditions (3) & (4): Manipulative content warning.

Explanation text for condition (4): "This post contains language that may be intended to make you feel **hope** by suggesting that a simple, inexpensive cure for cancer is available but is being ignored by pharmaceutical companies, implying a potential breakthrough in cancer treatment that is being withheld from the public."

Headline: Another medical use for pot: Healing broken bones

Summary: There's yet another use for marijuana: It may help to heal broken bones, according to a new study. Researchers found that cannabidiol - an element of marijuana that does not get people high - improved the healing process in rats with broken leg bones after eight weeks, according to a study published in the Journal of Bone and Mineral Research by Tel Aviv University and Hebrew University.

Post type: True + not manipulative.

Warning labels: No warning labels for any group.

Headline: Vegetarians tend to be slimmer and less extroverted than meat eaters, study finds

Summary: The less animal products someone consumes, the lower his body mass index on average and the less he tends to be extroverted. A connection with depressive moods as other studies had found could not be confirmed.

Post type: True + not manipulative.

Warning labels: No warning labels for any group.

Headline: Five-foot long tapeworm came 'wiggling out' of man's body after he ate sushi

Summary: A California man who ate sushi every day ended up with a 5-foot-plus long tapeworm inhabiting his body. The Fresno male went to the emergency room complaining of bloody diarrhea, according to Dr. Kenny Bahn, who shared the story of his patient on the podcast "This Won't Hurt A Bit."

Post type: True + manipulative. (*appeal to emotion: "anxiety"*)

Warning labels: Conditions (3) & (4): Manipulative content warning.

Explanation text for condition (4): "This post contains language that may be intended to make you feel anxiety by detailing a disturbing medical case involving a large tapeworm and linking it to a common food source, which may heighten concerns about the safety of eating sushi or salmon."

Headline: ESSENTIAL OIL DIFFUSERS POISONOUS FOR CATS!

Summary: Somebody posted this which is very interesting: This is my daughter's cat, Ernie. He has lived with us most of his 16 years. I unknowingly have been poisoning him since Christmas and feel the need to warn everyone who might be unaware of the toxicity of essential oils...

Post type: True + manipulative. (appeal to emotion: "fear")

Warning labels: Conditions (3) & (4): Manipulative content warning.

Explanation text for condition (4): "This post contains language that may be intended to make you feel fear by describing the serious health risks posed to cats by essential oil diffusers and the distressing effects on a specific pet, thereby creating concern about the safety of using such products around animals."

A.2 ChatGPT Prompt for Group (4): Manipulative text warning + explanation

The explanatory text for condition (4) was generated using ChatGPT (GPT-4) on July 31, 2024.

First, ChatGPT was prompted with the following instruction:

I will share articles and social media posts with you. I will inform you if the article/post has been determined (by another AI system) to contain emotionally manipulative language. In other words, whether the article/post contains appeals to emotion like fear, anger, hope, and anxiety, that are likely intended to manipulate readers into believing the article/post's content.

For reference, here is how we define each of those emotions:

fear: make people feel scared
 anger: make people angry
 hope: elicit hope or positive emotions
 anxiety: creates uncertainty of the future

I want you to tell me the following about each article:

Does the article indeed contain an appeal to emotion that could be manipulative? ("Yes" or "No")

If "Yes", write a 1-sentence warning explaining what emotions the article seems to be trying to stir. Please write the names of the emotions in bold. The sentence should start with: "This post contains language that may be intended to make you feel"...

Finally, if the article contains manipulative appeals to other emotions besides the four emotions listed earlier (fear, anger, hope, anxiety), please write a second warning that includes those emotions.

After this instruction, the following prompts were input for each of the posts deemed to contain emotional appeals based on Kamali et al.'s classification [25].

Here is an article that was determined to contain appeals to {EMOTION}, which may be manipulative.

{ARTICLE HEADLINE}

{ARTICLE CONTENT}

In these prompts, "{EMOTION}" was replaced with "hope," "fear," or "anxiety" based on the types of emotional appeal associated with each article. "{ARTICLE HEADLINE}" and "{ARTICLE CONTENT}" were replaced with the actual headline and content from each article.

The responses to those prompts indicated agreement with the emotional appeal associated with each article and included an explanation about what emotions the article seems to be attempting to evoke and how. These explanations were used in the warning labels for condition (4) with no alteration. See Appendix A.1 to see the full explanation texts.